11A 1 TECHNIQUES FOR MEASURING THE RATE OF REACTION

SPECIFICATION REFERENCE

11.1(i) 11.3(i) 11.3(ii) 11.3(iii) 11.3(iv) 11.3(v)

LEARNING OBJECTIVES

- Understand the term 'rate of reaction'.
- Select and justify a suitable experimental tecÜique to obtain rate data for a given reaction, including:
  - (i) titration
  - (ii) colorimetry
  - (iii) mass change
  - (iv) volume of gas evolved
  - (v) other suitable tecÜique(s) for a given reaction.

RATE OF REACTION

The **rate of a reaction** can be expressed in two ways:

(1) How the concentration of a product *increases* with time.

$$\text{rate} = \frac{\text{change in concentration of product}}{\text{time}}$$

(2) How the concentration of a reactant *decreases* with time.

$$\text{rate} = -\frac{\text{change in concentration of reactant}}{\text{time}}$$

The negative sign in the second expression shows that the concentration of the reactant is decreasing and therefore gives a positive value for the rate.

Rate is measured in units of concentration per unit time, and the most common units are mol dm^{-3} s^{-1}.

The expressions in calculus notation are:

$$\text{rate} = \frac{d[\text{product}]}{dt}$$

$$\text{rate} = -\frac{d[\text{reactant}]}{dt}$$

This **rate of reaction** is sometimes called the 'overall rate of reaction'.

TECHNIQUES FOR MEASURING THE RATE OF REACTION

Before investigating the rate of a particular reaction, it is necessary to know the overall equation, including state symbols, for the reaction so that we can decide what technique to use to follow the reaction.

There are various techniques available to use, such as:

1 measuring the volume of a gas evolved
2 measuring the change in mass of a reaction mixture
3 monitoring the change in intensity of colour of a reaction mixture (colorimetry)
4 measuring the change in concentration of a reactant or product using titration
5 measuring the change in pH of a solution
6 measuring the change in electrical conductivity of a reaction mixture.

The technique chosen to follow the reaction will depend on the nature of the reactants and products, as well as the conditions under which the reaction is carried out.

For example, the reaction between calcium carbonate and dilute hydrochloric acid,

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

could conveniently be followed by measuring the volume of gas (technique 1) given off at regular time intervals, or by measuring the change in mass of the reaction mixture with time (technique 2).

However, the reaction between propanone and iodine in aqueous solution,

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

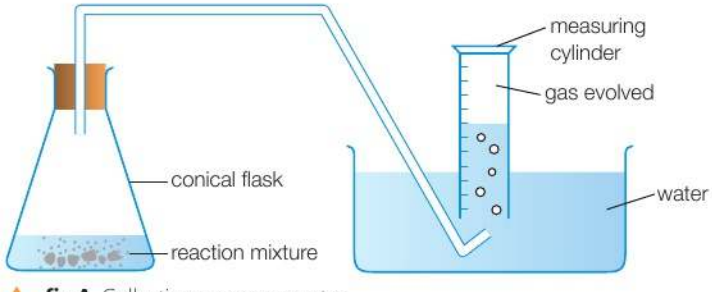
could not be followed by measuring the change in mass because all products of the reaction remain in solution. It would be possible, however, to follow the reaction by monitoring the decrease in intensity of colour of the reaction mixture, since $I_2(aq)$ is the only coloured **species** present (technique 3).

TECHNIQUE 1: MEASURING THE VOLUME OF A GAS EVOLVED

The two most common techniques for collecting and measuring the volume of a gas evolved during a reaction are:

1 collection over water into a measuring cylinder (**fig A**), and
2 collection using a gas syringe (**fig B**).

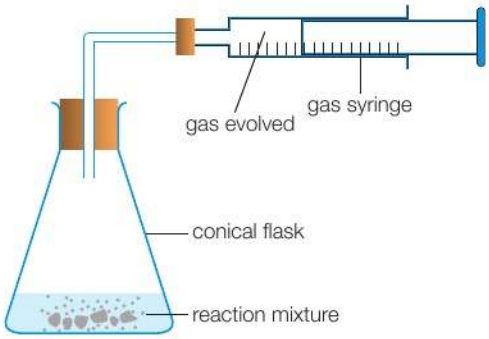
The technique chosen will depend partly on the level of precision required. The gas syringe has a greater degree of precision, but if a large volume of gas is being measured, the difference in the degree of measurement uncertainty becomes so small that either instrument is sufficiently precise.



▲ **fig A** Collecting a gas over water.

EXAM HINT

Reactions producing gases that are very soluble in water, such as sulfur dioxide, cannot use the gas collection over water tecÚique.



▲ **fig B** Collecting a gas in a gas syringe.

TECHNIQUE 2: MEASURING THE CHANGE IN MASS OF A REACTION MIXTURE

This is another technique applicable to reactions in which a gas is evolved.

The reaction flask and contents are placed on a digital balance and the decrease in mass is measured as the reaction proceeds (**fig C**).

This technique is most precise when the gas given off has a relatively high density, such as with carbon dioxide. With a low-density (i.e. low relative molecular mass) gas such as hydrogen, the mass changes are so small that the measurement uncertainties become significant.



▲ **fig C** Cotton wool is placed in the neck of the flask to prevent the loss of liquid spray.

TECHNIQUE 3: MONITORING A COLOUR CHANGE (COLORIMETRY)

Colour change can sometimes be monitored using observation only. However, using a colorimeter gives more precise results (**fig D**). A colorimeter can detect far more subtle changes than the human eye can observe, and provides a quantitative (rather than a subjective) measurement.



▲ **fig D** A colorimeter.

TECHNIQUE 4: ANALYSIS BY TITRATION

This technique involves using a pipette to remove small samples (aliquots) from a reaction mixture at regular intervals. The reaction in the aliquot can either be stopped by adding another substance to it or slowed down to almost zero by immersing it in an ice bath. The aliquot is then titrated to determine the concentration of a reactant or product species.

The process of stopping or slowing down the reaction in an aliquot is known as 'quenching'.

For example, if the reaction involves an acid, the aliquot, after quenching, could be titrated against a standard solution of sodium hydroxide to determine the concentration of the acid. This technique is used to investigate the reaction between iodine and propanone, which is catalysed by acid. Sodium hydrogen carbonate is added to the aliquot to remove the acid catalyst and, as a result, effectively stops the reaction. The remaining iodine is then titrated against a standard solution of sodium thiosulfate (**fig E**).

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$



▲ **fig E** Titrating iodine against sodium thiosulfate.

TECHNIQUE 5: MEASURING THE ELECTRICAL CONDUCTIVITY

If the total number, or type, of ions in solution changes during a reaction, it might be possible to follow the reaction by measuring changes in the electrical conductivity of the solution using a conductivity meter. For example, it could be used to follow this reaction:

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

TECHNIQUE 6: MEASURING ANY OTHER PHYSICAL PROPERTY THAT SHOWS A SIGNIFICANT CHANGE

Possible physical properties that have not already been mentioned include changes in the volume of liquid ('dilatometry'), chirality and refractive index.

CHECKPOINT

1. State suitable tecÚiques to collect rate data for each of the following reactions.

(a) Magnesium with dilute sulfuric acid:

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

(b) Ethyl ethanoate with sodium hydroxide:

$$CH_3COOCH_2CH_3(l) + OH^-(aq) \rightarrow CH_3COO^-(aq) + CH_3CH_2OH(aq)$$

(c) Hydrogen gas with iodine gas:

$$H_2(g) + I_2(g) \rightarrow 2HI(g)$$

SKILLS CREATIVITY

2. Why would the tecÚique of measuring the change in mass of a reaction vessel and contents not work well in the reaction between magnesium and dilute sulfuric acid?

3. The reaction between calcium carbonate and dilute hydrochloric acid can be followed by collecting and measuring the volume of gas produced. The gas could be collected over water in a measuring cylinder or in a gas syringe. Which tecÚique would be the more suitable for this reaction? Explain your answer.

SUBJECT VOCABULARY

(overall) **rate of reaction** the change in concentration of a species divided by the time it takes for the change to occur. All reaction rates are positive

(chemical) **species** an atom, a molecule or an ion that is taking part in a chemical reaction

11A 2 RATE EQUATIONS, RATE CONSTANTS AND ORDERS OF REACTION

SPECIFICATION REFERENCE
11.1(ii) 11.1(iii) 11.1(iv) 11.1(v) 11.1(vii)

LEARNING OBJECTIVES

- Understand the terms:
  - (i) rate equation, rate = $k[A]^m[B]^n$, where m and n are 0, 1 or 2
  - (ii) order with respect to a substance in a rate equation
  - (iii) overall order of a reaction
  - (iv) rate constant
  - (v) rate-determining step.

RATE EQUATION

WHAT IS A RATE EQUATION?

The usual relationship between the rate of reaction and the concentration of a reactant is that the rate of reaction is directly proportional to the concentration. In other words, as the concentration is doubled, the rate of reaction doubles.

Unfortunately, this is not always the case. Sometimes the rate will double, but sometimes it will increase by a factor of four. In some cases, the rate of reaction does not increase at all, or it increases in an unexpected way.

Let us consider the simple relationship where the rate is directly proportional to the concentration of a reactant, say A. We can represent this by the expression:

$$\text{rate} \propto [A]$$

or:

$$\text{rate} = k[A]$$

where k is the proportionality constant.

This is called the *first order* **rate equation**. The constant k is called the *rate constant*.

- Every reaction has its own particular rate equation and its own rate constant.
- Rate constants will only change their value with a change in temperature.

Other common rate equations with respect to an individual reactant are:

second order rate equation: $\text{rate} = k[A]^2$

zero order rate equation: $\text{rate} = k[A]^0$ or $\text{rate} = k$

Zero order reactions do not occur very often, and it might be difficult at this stage for you to imagine why they should occur at all. However, you will find out shortly why they can occur.

If two or more reactants are involved, then it is possible to have a third order rate equation:

third order rate equation: $\text{rate} = k[A]^2[B]$

THE UNITS OF RATE CONSTANTS

Table A shows the units for rate constants, using mol dm^{-3} as the unit of concentration and seconds as the unit of time. You find the units by rearranging the rate equation.

ORDER	UNIT
Zero	mol dm^{-3} s^{-1}
First	s^{-1}
Second	dm^3 mol^{-1} s^{-1}
Third	dm^6 mol^{-2} s^{-1}

table A

EXAM HINT

The values for orders of reaction in rate equations will only ever be 0, 1 or 2 at International A Level.

The units are obtained by rearranging the rate equation. For example, for a second order reaction:

$$k = \frac{\text{rate}}{[\text{A}]^2}$$

Inserting the units we obtain:

$$\frac{\text{mol dm}^{-3}\,\text{s}^{-1}}{\text{mol dm}^{-3} \times \text{mol dm}^{-3}}$$

This cancels down to:

$$\frac{\text{s}^{-1}}{\text{mol dm}^{-3}}$$

which equates to $\text{dm}^3\,\text{mol}^{-1}\,\text{s}^{-1}$.

The majority of reactions involve two or more reactants. If we call the reactants A, B and C, then the reaction may be first **order** with respect to A, first order with respect to B and second order with respect to C. The *overall* rate equation will be:

$$\text{rate} = k[\text{A}][\text{B}][\text{C}]^2$$

and the **overall order** of the reaction is four (1 + 1 + 2). Note that you are adding the powers.

For a general reaction in which the orders are m, n and p, we have:

$$\text{rate} = k[\text{A}]^m[\text{B}]^n[\text{C}]^p$$

The overall order of the reaction is $m + n + p$.

REACTION MECHANISMS

Many reactions can be represented by a stoichiometric equation containing many reactant particles. For example, the reaction between manganate(VII) ions and iron(II) ions in acidic solution can be represented by:

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

If this reaction actually proceeded in a single step, then the reaction would be very slow indeed. The probability of 14 particles simultaneously colliding, all with the correct orientation and energy, is so small that we can say it is effectively zero. The reaction is, however, very fast indeed even at room temperature. It must, therefore, proceed via a series of steps, all of which follow on quickly from one another. It is important to recognise that a reaction involving simultaneous collision of more than two particles is very rare.

The orders of reaction of the individual reactants can help us to suggest a possible mechanism for a reaction. The mechanism cannot be inferred from the stoichiometric equation, because the mathematical relationship between the rate of reaction and the concentration of reactants (i.e. the orders of reaction) can only be determined *through experiments*.

Consider the reaction:

$$\text{A} + \text{B} + \text{C} \rightarrow \text{D} + \text{E}$$

for which the experimentally determined rate equation is:

$$\text{rate} = k[\text{A}][\text{B}]$$

This could mean that C was present in such a large excess that changes in its concentration were negligible and therefore had no measurable effect on the rate of reaction.

If, however, this is not the case, and changes in [C] really do not have any effect on the overall rate of reaction, then a different explanation must be sought for why [C] does not appear in the rate equation.

In this case, there must be a step involving a reaction between A and B that has an effect on the rate of reaction. There must also be another step in which C reacts, but this reaction has *no effect* on the overall rate of reaction. This could be explained by assuming that the reaction between A and B takes place *before* C has a chance to react, and that the reaction between A and B is significantly *slower* than the reaction involving C. If this were the case, then the mechanism for the reaction could be:

Step 1:	$\text{A} + \text{B} \rightarrow \text{Z}$	SLOW
Step 2:	$\text{Z} + \text{C} \rightarrow \text{D} + \text{E}$	FAST

Since only Step 1 is **rate-determining**, then only changes in [A] and [B] will affect the overall rate of reaction. Changes in the rate at which Step 2 occurs, owing to changes in [C], will be negligible.

Important points to remember are that:

- *The slowest step in a reaction determines the overall rate of the reaction.*
- The slowest step is known as the *rate-determining step* of the reaction.

We will return to the concept of reaction mechanisms, and consider them in much more detail, in **Section 11A.4**.

WORKED EXAMPLE

A useful way of visualising the idea of a rate-determining step is to imagine that three students are arranging some sheets of notes into sets (**fig A**).

Step 1: the notes are arranged into ten piles and the first student collects a sheet from each of the piles.

Step 2: the second student takes the set of ten papers and shuffles them so that they are tidy and ready for stapling.

Step 3: the third student staples the set of notes together.

The overall rate of this process (i.e. the rate at which the final sets of notes are prepared) depends on the rate of Step 1, the collecting of the sheets of notes, since this is by far the slowest step.

It does not matter, within reason, how quickly the tidying up for stapling is done. For most of the time the second and third students will be doing nothing while they wait for the first student to collect the sheets. The mechanism of the process is therefore:

Step 1:	Student 1 collects sheets	SLOW
Step 2:	Student 2 tidies set of sheets	FAST
Step 3:	Student 3 staples set of sheets	FAST



▲ **fig A** Analogy for rate-determining step.

LEARNING TIP

All reactant species involved either in, or before, the rate-determining step have an effect on the rate of the reaction and will appear in the rate equation.

CHECKPOINT

1. The rate equation for the reaction between peroxydisulfate ions and iodide ions is:

rate = $k[S_2O_8^{2-}][I^-]$

(a) What is the order of reaction with respect to (i) peroxydisulfate ions and (ii) iodide ions?

(b) What is the overall order of reaction?

2. The rate equation for the reaction between P and Q is:

rate = $k[P][Q]^2$

What will be the increase in rate if:

(a) [P] is doubled, while [Q] is kept constant?

(b) [Q] is doubled, while [P] is kept constant?

(c) [P] and [Q] are both doubled?

3. In the reaction between R, S and T:

- when the concentration of R is doubled, the rate increases by four times
- when the concentration of S is doubled, the rate does not change
- when the concentration of T is doubled, the rate doubles.

(a) Deduce the orders of reaction with respect to R, S and T.

SKILLS PROBLEM-SOLVING

(b) Write the rate equation for the reaction.

(c) What is the overall order of reaction?

SUBJECT VOCABULARY

rate equation an equation expressing the mathematical relationship between the rate of reaction and the concentrations of the reactants

order (of a reactant species) the power to which the concentration of the species is raised in the rate equation

overall order (of a reaction) the sum of all the individual orders

rate-determining step (of a reaction) the slowest step in the mechanism for the reaction

11A 3 DETERMINING ORDERS OF REACTION

SPECIFICATION REFERENCE

11.1(ii) 11.1(vi) 11.2 11.4(i) 11.4(ii) 11.5(i) 11.5(ii) 11.5(iii)

LEARNING OBJECTIVES

- Understand the terms:
  (i) rate equation, rate = $k[A]^m[B]^n$, where m and n are 0, 1 or 2
  (ii) half-life.
- Be able to calculate the half-life of a reaction, using data from a suitable graph, and identify a reaction with a constant half-life as being first order.
- Understand experiments that can be used to investigate reaction rates by:
  (i) an initial-rate method
  (ii) a continuous monitoring method.
- Be able to deduce the order (0, 1 or 2) with respect to a substance in a rate equation, using data from:
  (i) a concentration–time graph
  (ii) a rate–concentration graph
  (iii) an initial-rate method.

HOW CAN WE DETERMINE THE RATE EQUATION?

This question can equally be phrased 'How can we determine the order of reaction with respect to each reactant?'

There are two methods for determining orders of reaction. They are both experimental. Indeed, orders of reaction can only be determined by experiment.

The first is sometimes called the 'continuous method'.

The second is sometimes called the 'initial-rate method'.

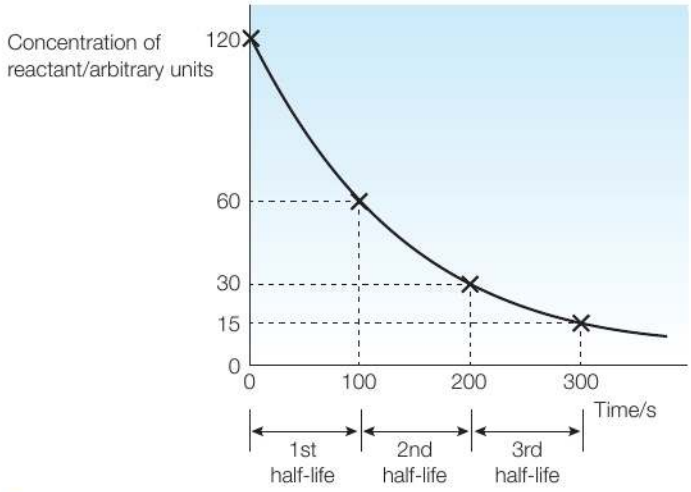
THE CONTINUOUS METHOD

In this method, *one* reaction mixture is made up and samples of the reaction mixture are withdrawn at regular time intervals. The reaction in the sample is stopped, if necessary, by quenching. The concentration of the reactant is then determined by an appropriate experimental technique, such as titration.

- The first step is to draw a 'concentration–time' graph.
- The second step is to find out the **half-life** for the reaction at different concentrations.

If the half-life has a constant value, then the reaction is first order with respect to the reactant.

The 1st half-life, for the change in concentration from 120 to 60 units, is 100 s (**fig A**).



▲ **fig A** A graph of concentration of reactant against time for a first order reaction.

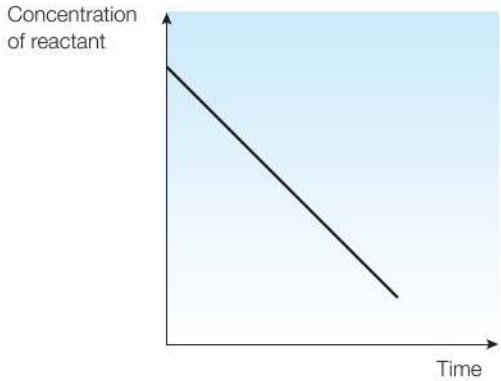
The 2nd half-life, for the change in concentration from 60 to 30 units, is 100 s.

The 3rd half-life, for the change in concentration from 30 to 15 units, is 100 s.

Since all three half-lives are the same, the reaction is first order with respect to the concentration of the reactant plotted.

If the half-life doubles as the reaction proceeds, then the reaction is second order.

If the graph is a straight line with a negative gradient, then the rate of reaction is constant no matter what the concentration of the reactant (**fig B**). In other words, the reaction is zero order with respect to the reactant.



▲ **fig B** A graph of concentration against time for a zero order reaction.

A typical set of results is shown in **table A** for the reaction:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

TIME/min	0	10	20	30	40	50	60	70	80	90
$[N_2O_5]/10^{-3}$ mol dm^{-3}	22.90	16.27	12.29	9.35	6.89	4.88	3.68	2.74	2.16	1.85

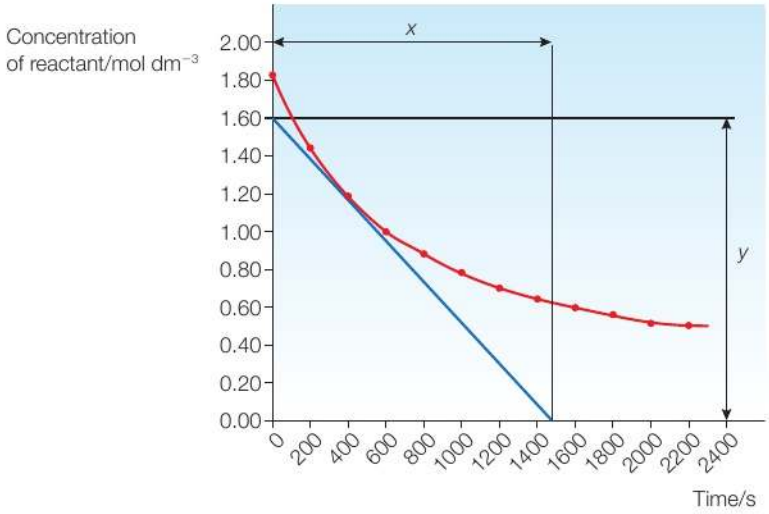
table A

If you plot a graph of $[N_2O_5]$ against time using the above data, you will find that the line is curved and the half-life is constant; as a result, the reaction is first order with respect to N_2O_5.

CALCULATING RATE FROM A CONCENTRATION–TIME GRAPH

The rate of reaction at any given time can be determined from a concentration–time or volume–time graph by drawing a tangent to the curve at the given time and calculating the gradient of the tangent.

Fig C shows the change in concentration of a reactant with time.



▲ **fig C** Determining an instantaneous rate of reaction from a graph of concentration of reactant against time.

A tangent to the curve has been drawn at time = 400 s.

To find the rate at this time point, draw as large a triangle as possible and then measure x and y.

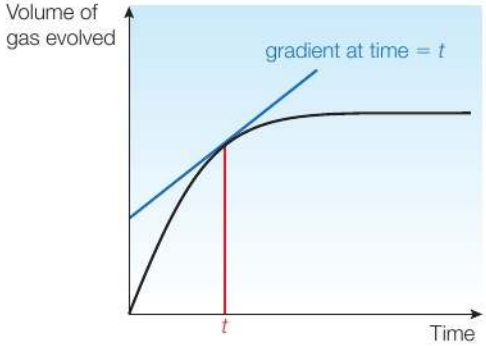
$x = 1470$ s and $y = -1.60$ mol dm^{-3}.

$$\text{rate} = -\frac{\text{change in concentration of reactant}}{\text{time}}$$

$$= -\frac{-1.60 \text{ mol dm}^{-3}}{1470 \text{ s}} = 1.09 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$$

CALCULATING RATE FROM A VOLUME–TIME GRAPH

The procedure is exactly the same for a volume–time graph for a gas evolved (**Section 11A.1**). This time the curve will slope upwards not downwards (**fig D**).



▲ **fig D** Determining an instantaneous rate of reaction from a graph of volume of gas evolved against time.

The rate of reaction obtained in this way is sometimes called the **instantaneous reaction rate**.

THE INITIAL-RATE METHOD FOR DETERMINING THE RATE EQUATION

In this method, *several* reaction mixtures are made up and the initial rate (i.e. the time taken for a fixed amount of reactant to be used up or for a fixed amount of product to be formed) is measured. From these times, it is then possible to calculate the mathematical relationship between the rate of the reaction and the concentration of the reactant.

A typical set of results is shown in **table B** for the reaction:

$$A + B \rightarrow C$$

EXPERIMENT	[A]/mol dm^{-3}	[B]/mol dm^{-3}	INITIAL RATE OF FORMATION OF C/mol dm^{-3} s^{-1}
1	0.1	0.1	0.02
2	0.1	0.2	0.04
3	0.2	0.1	0.04
4	0.2	0.2	0.08

table B

If we now look at experiments 1 and 3, we can see that [A] has doubled while [B] has remained constant. The rate of reaction has also doubled. This indicates that the reaction is first order with respect to A and, as a result, we can write:

$$\text{rate} \propto [A]$$

If we now look at experiments 1 and 2, or experiments 3 and 4, we can see that [B] has doubled while [A] has remained constant. The initial rate of reaction has also doubled. This indicates that the rate of reaction is directly proportional to [B]. That is, the reaction is first order with respect to B. As a result, we can write:

$$\text{rate} \propto [B]$$

The overall order of the reaction is two, and the complete rate law (rate equation) is:

$$\text{rate} = k[A][B]$$

Note that the numbers in **table B** were deliberately kept simple to illustrate a point. Real experimental results rarely fit such an exact pattern, and you may have to look for the nearest whole numbers to obtain orders.

DETERMINING ORDER FROM A RATE–CONCENTRATION GRAPH

The method described above to determine initial rates, with changing concentrations, results in only approximate values for the initial rates. However, since most orders of reaction with respect to individual reactants have integer values, this approximation is usually acceptable.

In an initial rates experiment, the time measured is that for a fixed amount of product to be formed, or that for a fixed amount of reactant to be used up. Since the amount of product formed, or reactant used up, is kept constant, the initial rate of reaction is proportional to the reciprocal of the time, t, measured. That is:

$$\text{rate} \propto \frac{1}{t}$$

This assumes that the rate is constant for the whole time period, t. However, this is not true because as soon as the reaction starts the rate begins to decrease. The rate calculated from the expression is, therefore, the *mean (average)* rate over time, t, and not the *true* initial rate of reaction. The approximation becomes poorer with larger values of t. However, the approximation is good enough to determine integer order.

It might be possible to determine orders of reaction from just a few measurements. However, it is usual to record a range of results and plot a graph of $\frac{1}{t}$ against concentration of reactant.

The shape of the graph in **table C** indicates the order of reaction.

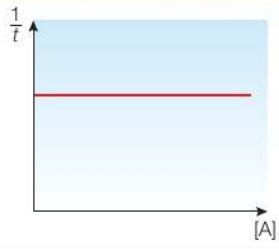
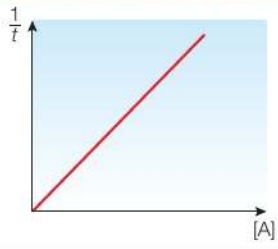
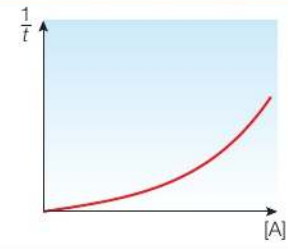
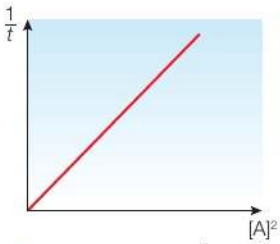
ZERO ORDER WITH RESPECT TO A $1/t$ ($\propto$RATE) = $k[A]^0$	FIRST ORDER WITH RESPECT TO A $1/t$ ($\propto$RATE) = $k[A]^1$	SECOND ORDER WITH RESPECT TO A $1/t$ ($\propto$RATE) = $k[A]^2$
		

table C

It is impossible to determine directly by sight from a rate–concentration graph that the reaction is second order. If the graph is a curve as shown, then it is necessary to then plot $\frac{1}{t}$ against $[A]^2$. If this produces a straight line passing through the origin, then the reaction is second order with respect to A.



▲ **fig E** A graph of $\frac{1}{t}$ against the square of the concentration of a reactant for a second order reaction.

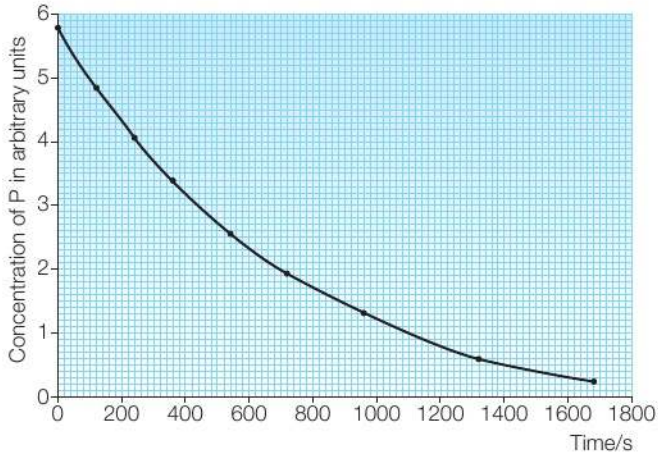
EXAM HINT

Be careful to check the units on the axes of graphs. Graphs with the same shape will have different meanings if there are different units on the axes!

CHECKPOINT

SKILLS INTERPRETATION

1. A compound P decomposes when heated. The graph shows the change in concentration when a sample of P is heated.



(a) State what is meant by the term *half-life of reaction*.

(b) Use the graph to show that the decomposition of P is a first order reaction.

(c) Explain the effect on the half-life of doubling the initial concentration of P.

(d) Calculate the rate constant, k, for this reaction using the following expression:

$$k = \frac{0.693}{\text{half-life}}$$

Include units in your answer.

(e) Write the rate equation for this reaction.

(f) (i) Use the graph to determine the concentration of P at 800 s.

(ii) Use the rate equation from (e) and your answers to (d) and (f)(i) to calculate the rate of reaction at 800 s. Include units in your answer.

(g) Describe how could you determine the reaction rate at 800 s directly from the graph.

SKILLS INTERPRETATION

2. The equation for the reaction between bromide ions and bromate(V) ions in acidified aqueous solution is:

$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$

The table shows the results of four experiments carried out using different concentrations of the three reactants.

EXPERIMENT	$[Br^-(aq)]$/ mol dm^{-3}	$[BrO_3^-(aq)]$/ mol dm^{-3}	$[H^+(aq)]$/ mol dm^{-3}	INITIAL RATE/ mol dm^{-3} s^{-1}
1	0.10	0.10	0.10	1.2×10^{-3}
2	0.10	0.20	0.10	2.4×10^{-3}
3	0.30	0.10	0.10	3.6×10^{-3}
4	0.10	0.20	0.20	9.6×10^{-3}

(a) Deduce the order of reaction with respect to:

(i) Br^-

(ii) BrO_3^-

(iii) H^+

(b) Write the rate equation for the reaction.

(c) Using the results from experiment 1, calculate the rate constant, k, for the reaction. Include units in your answer.

SUBJECT VOCABULARY

half-life (of a reaction) the time taken for the concentration of the reactant to fall to one-half of its initial value

instantaneous reaction rate the gradient of a tangent drawn to the line of the graph of concentration against time. The instantaneous rate varies as the reaction proceeds (except for a zero order reaction)

11A 4 RATE EQUATIONS AND MECHANISMS

SPECIFICATION REFERENCE

11.6	11.7	11.8
11.9	CP9A	CP9B

LEARNING OBJECTIVES

- Deduce the rate-determining step from a rate equation and vice versa.
- Deduce a reaction mechanism, using knowledge of the rate equation and the stoichiometric equation for a reaction.
- Understand that knowledge of the rate equations for the hydrolysis of halogenoalkanes can be used to provide evidence for S_N1 or S_N2 mechanisms for tertiary and primary halogenoalkane hydrolysis.
- Understand how to:
  - (i) obtain data to calculate the order with respect to the reactants (and the hydrogen ion) in the acid-catalysed iodination of propanone
  - (ii) use these data to make predictions about species involved in the rate-determining step
  - (iii) deduce a possible mechanism for the reaction.

REACTION MECHANISMS

You will remember from **Book 1** that the basic view as to how a reaction takes place at a particulate level is that particles (atoms, molecules, ions or radicals) first have to collide in the correct orientation and with sufficient energy for products to be formed.

For the following reaction:

$$P + Q \rightarrow \text{products}$$

we expect the rate law to be:

$$\text{rate} = k[P][Q]$$

In other words, we might expect the reaction to be first order with respect to each reactant and second order overall.

ELEMENTARY REACTIONS

A reaction taking place in this manner (a single collision between the two reactant particles) is described as being *elementary*.

If we know that a reaction is elementary, then we can deduce the rate law directly from the stoichiometric equation. For example, the following reaction is known to be elementary:

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$$

so the rate equation is:

$$\text{rate} = k[NO][O_3]$$

The reaction takes place when a molecule of NO collides with a molecule of O_3.

If the reaction is not elementary, it is *not* possible to deduce the rate equation by simply looking at the stoichiometric equation for the reaction.

For example, the decomposition of dinitrogen pentoxide into nitrogen dioxide and oxygen is first order with respect to dinitrogen pentoxide, not second.

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

The experimentally determined rate equation is:

$$\text{rate} = k[N_2O_5]$$

A reaction that is not elementary takes place via a series of interconnected elementary reactions that are collectively called the *mechanism* for the reaction. You will have already come across a number of such mechanisms in your study of organic chemistry. For example, the radical substitution reaction between methane and chlorine to form chloromethane (CH_3Cl) is thought to have the following mechanism:

Step 1: $Cl_2 \rightarrow 2Cl\bullet$

Step 2: $Cl\bullet + CH_4 \rightarrow HCl + \bullet CH_3$

Step 3: $Cl_2 + \bullet CH_3 \rightarrow Cl\bullet + CH_3Cl$

The overall stoichiometric equation for the reaction is:

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$$

The species $Cl\bullet$ and $\bullet CH_3$ are called *intermediates*. They do not appear in the overall equation for the reaction, but are involved in reactions that ultimately result in the reactants being converted into the products.

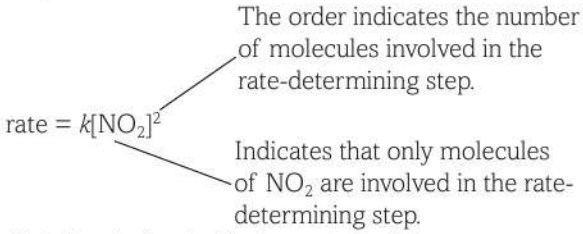
If the experimentally determined rate equation does not match the overall stoichiometry, then it is almost certain that the reaction is not elementary. For example, the rate equation for the following reaction:

$$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$$

is:

$$\text{rate} = k[NO_2]^2$$

This suggests that only molecules of NO_2 are involved in the rate-determining step, and that *two* molecules of NO_2 are involved in this step.



▲ **fig A** Molecules involved in the rate-determining step.

Using our knowledge of molecules that do exist, two possible rate-determining steps are:

1 $NO_2 + NO_2 \rightarrow N_2O_4$, or

2 $NO_2 + NO_2 \rightarrow 2NO + O_2$

Both of these reactions are equally valid, and we have no way of knowing, without carrying out further investigations, which is the more likely to be taking place.

We also have no way of knowing what is involved in the remaining steps, but we do know that the sum of all the steps must add up to the overall stoichiometric equation.

Using the second of the two possible rate-determining steps, the following mechanism is consistent with the data:

$NO_2 + NO_2 \rightarrow 2NO + O_2$ SLOW

$CO + O_2 \rightarrow CO_2 + O$ FAST

$NO + O \rightarrow NO_2$ FAST

This shows how the particles in the proposed mechanism cancel to produce the overall equation:

$$\begin{array}{rcl} NO_2 + \cancel{NO_2} & \rightarrow & \cancel{NO} + NO + \cancel{O_2} \\ CO + \cancel{O_2} & \rightarrow & CO_2 + \cancel{O} \\ \cancel{NO} + \cancel{O} & \rightarrow & \cancel{NO_2} \\ \hline NO_2 + CO & \rightarrow & NO + CO_2 \end{array}$$

As it happens, further investigations into this reaction have identified the mechanism as:

$NO_2 + NO_2 \rightarrow NO_3 + NO$ SLOW

$NO_3 + CO \rightarrow NO_2 + CO_2$ FAST

This is not what we would immediately suspect as the mechanism because the existence of NO_3 is not something with which we would be familiar.

Even if the experimentally determined rate equation is simple, it does not necessarily follow that the reaction proceeds in a single elementary step. For example, the rate expression for the following gas-phase reaction:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

is:

$$\text{rate} = k[N_2O_5]$$

Yet, the reaction is thought to involve several steps and a number of intermediates.

ALKALINE HYDROLYSIS OF HALOGENOALKANES

The hydrolysis of halogenoalkanes by hydroxide ions is a reaction we introduced in **Topic 10 (Book 1: IAS)**. The hydroxide ion acts as a nucleophile and replaces the halogen in the halogenoalkanes. The reaction, therefore, can also be described as *nucleophilic substitution*.

HYDROLYSIS OF A TERTIARY HALOGENOALKANE

The equation for the alkaline hydrolysis of 2-chloromethylpropane is:

$$(CH_3)_3CCl + OH^- \rightarrow (CH_3)_3COH + Cl^-$$

The experimentally determined rate equation for this reaction is:

$$\text{rate} = k[(CH_3)_3CCl]$$

The reaction is first order with respect to 2-chloromethylpropane, but zero order with respect to the hydroxide ion.

The sensible conclusion to reach is that the 2-chloromethylpropane undergoes slow ionisation as the rate-determining step. This is then followed by a very fast step involving attack by the hydroxide ion on the carbocation formed in Step 1.

Step 1: $(CH_3)_3CCl \rightarrow (CH_3)_3C^+ + Cl^-$ SLOW

Step 2: $(CH_3)_3C^+ + OH^- \rightarrow (CH_3)_3COH$ FAST

This type of mechanism is named S_N1, i.e. *S*ubstitution *N*ucleophilic *uni*molecular.

The rate-determining step is said to be *uni*molecular because there is only *one* reactant particle present, $(CH_3)_3CCl$.

The carbocation, $(CH_3)_3C^+$, formed in Step 1 is an *intermediate* (see **Section 11A.6**).

HYDROLYSIS OF A PRIMARY HALOGENOALKANE

The equation for the alkaline hydrolysis of 1-chlorobutane is:

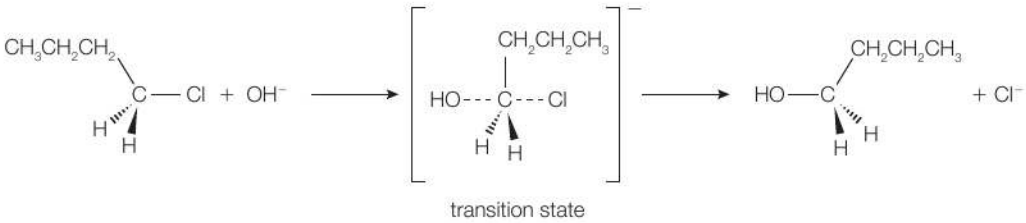
$$CH_3CH_2CH_2CH_2Cl + OH^- \rightarrow CH_3CH_2CH_2CH_2OH + Cl^-$$

The experimentally determined rate equation for this reaction is:

$$\text{rate} = k[CH_3CH_2CH_2CH_2Cl][OH^-]$$

This time the reaction is first order with respect to each reactant, so it is reasonable to suggest that one particle of each reactant is present in the rate-determining step of the mechanism.

The accepted mechanism for the reaction is:



▲ **fig B** S_N2 mechanism of the alkaline hydrolysis of 1-chlorobutane.

This type of mechanism is named S_N2, i.e. *S*ubstitution *N*ucleophilic *bi*molecular.

The rate-determining step is *bi*molecular because there are *two* reactant particles present.

It is a continuous single one-step reaction. The complex shown in square brackets is not an intermediate (like the carbocation formed in the S_N1 mechanism), but is a *transition state* (see **Section 11A.6**).

PRACTICAL SKILLS CP9A

A study of the acid-catalysed iodination of propanone

The reaction between propanone and iodine in aqueous solution may be acid catalysed:

$$I_2(aq) + CH_3COCH_3(aq) + H^+(aq) \rightarrow CH_3COCH_2I(aq) + 2H^+(aq) + I^-(aq)$$

The influence of the iodine on the reaction rate may be studied if the concentrations of propanone and hydrogen ions effectively remain constant during the reaction. This is achieved by using a *large excess* of both propanone and acid in the original reaction mixture. You will investigate this in **CP9A Following the rate of iodine–propanone reaction by a titrimetric method**.

Procedure

1 Mix 25 cm^3 of 1 mol dm^{-3} aqueous propanone with 25 cm^3 of 1 mol dm^{-3} sulfuric acid.

2 Start the stop clock *the moment you add* 50 cm^3 of 0.02 mol dm^{-3} iodine solution. Shake well.

3 Using a pipette, withdraw a 10 cm^3 sample and place it in a conical flask. Stop the reaction by adding a 'spatula-measure' of sodium hydrogen carbonate. Note the exact time at which the sodium hydrogen carbonate is added.

4 Titrate the remaining iodine present in the sample with 0.01 mol dm^{-3} sodium thiosulfate(VI) solution, using starch indicator.

5 Withdraw further 10 cm^3 samples at suitable time intervals (approx. 5 to 7 minutes) and treat them similarly, always noting the exact time at which the sodium hydrogen carbonate is added. Waste should be contained in a fume cupboard.

Treatment of results

The Lab Book can give additional guidance in the treatment and analysis of results.

- Plot a graph of titre against time. (The titre is proportional to the concentration of iodine.)
- Deduce from the graph the order of reaction with respect to iodine.

Analysis of results

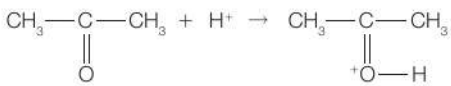
The graph produced shows that the reaction is zero order with respect to iodine.

Similar experiments show that the reaction is first order with respect to both propanone and hydrogen ions.

This gives us the following rate equation:

$$\text{rate} = k[CH_3COCH_3][H^+]$$

This would suggest the following reaction as the first step of the reaction:

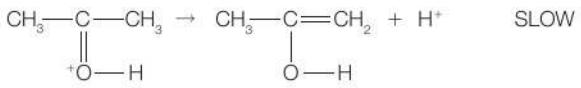


Similar reactions in other mechanisms are very fast, so this reaction is unlikely to be the rate-determining step of this reaction.

EXAM HINT

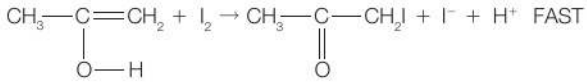
Examiners may ask you to **evaluate** different methods for investigating rates of reaction. One limitation of the method described here is the time delay between withdrawing a reaction sample by pipette and quenching it with sodium hydrogen carbonate.

The second step probably controls the rate of the reaction and produces H^+ that, being a catalyst, is not used up in the reaction:



This rearrangement is likely to be very slow and hence is probably the rate-determining step of the reaction.

Iodine can now react in a fast step as follows:



Testing the mechanism

We have a proposed mechanism for the reaction that is consistent with the kinetic data obtained from experiment. This is not the same as saying that the mechanism is the correct one, or indeed the most likely one.

We now need to carry out further experiments to confirm, or deny, our proposed mechanism. Here are three tecÚiques that can be employed in this case.

- *Use a wider range of concentrations*

  The experimentally determined rate equation may hold over only a limited range of concentrations. For example, the mechanism we have proposed for the iodination of propanone predicts that, at very low concentrations of iodine, the order of reaction with respect to iodine will change from zero to first. This is because the rate of the final step in the mechanism is given by the expression:

  Rate = $k[I_2][CH_3C(OH)=CH_2]$

  and so the rate of the step will decrease as $[I_2]$ decreases. This has been shown to be the case, so supports the proposed mechanism.

- *Use instrumental analysis*

  This may be able to detect the presence of intermediates that have been proposed. Nuclear magnetic resonance can be used to show that acidified propanone contains about one molecule in a million in the form $CH_3C(OH)=CH_2$. This is an intermediate stated in the proposed mechanism.

- *Carry out the reaction with deuterated propanone, CD_3COCD_3*

  Deuterium ^2_1H (or simply D) behaves slightly differently to hydrogen ^1_1H (or simply H).

  A C–D bond is slightly stronger, and therefore harder to break, than a C–H bond. If this is the bond that breaks in the rate-determining step, as suggested by our proposed mechanism, then the iodination of CD_3COCD_3 will be slower than that of CH_3COCH_3. This is found to be the case.

CHECKPOINT

1. The equation for the reaction between ethanal, CH_3CHO, and hydrogen cyanide, HCN, is:

$CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$

Two mechanisms that have been proposed for this reaction are:

Mechanism 1

Step 1: $CH_3CHO + H^+ \rightarrow [CH_3CHOH]^+$

Step 2: $[CH_3CHOH]^+ + CN^- \rightarrow CH_3CH(OH)CN$

Mechanism 2

Step 1: $CH_3CHO + CN^- \rightarrow [CH_3CHOCN]^-$

Step 2: $[CH_3CHOCN]^- + H^+ \rightarrow CH_3CH(OH)CN$

The rate equation for the reaction is:

rate = $k[CH_3CHO][CN^-][H^+]^0$

(a) Explain which of the two mechanisms is consistent with the rate equation.

(b) Which step in this mechanism is the rate-determining step?

2. The equation for the reaction between hydrogen and nitrogen monoxide is:

$2H_2(g) + 2NO(g) \rightarrow 2H_2O(g) + N_2(g)$

The rate equation for the reaction is:

rate = $k[H_2][NO]^2$

A proposed mechanism for this reaction is:

Step 1:	$2NO(g) \rightarrow N_2O_2(g)$	FAST
Step 2:	$N_2O_2(g) + H_2(g) \rightarrow H_2O(g) + N_2O(g)$	SLOW
Step 3:	$N_2O(g) + H_2(g) \rightarrow N_2(g) + H_2O(g)$	FAST

Is this mechanism consistent with the rate equation? Explain your answer.

3. Assume the following proposed reaction mechanism is correct:

$Cl_2 \rightarrow 2Cl\bullet$	SLOW
$H_2 + Cl\bullet \rightarrow HCl + H\bullet$	FAST
$H\bullet + Cl\bullet \rightarrow HCl$	FAST

(a) Write the overall equation for the reaction.

(b) Write a rate equation for the reaction that is consistent with the mechanism.

(c) What would be the effect on the rate of reaction of doubling the concentration of Cl_2?

(d) What would be the effect on the rate of reaction of doubling the concentration of H_2?

4. The nucleophilic substitution reaction between equimolar quantities of CH_3Cl and OH^- is second order overall. However, if the reaction is carried out using a large excess of OH^-, the reaction becomes first order overall. Suggest an explanation for these observations.

5. Bromine can be formed by the oxidation of hydrogen bromide with oxygen.

A proposed mechanism for this reaction is:

Step 1: $HBr + O_2 \rightarrow HBrO_2$

Step 2: $HBrO_2 + HBr \rightarrow 2HBrO$

Step 3: $HBrO + HBr \rightarrow Br_2 + H_2O$

Step 4: $HBrO + HBr \rightarrow Br_2 + H_2O$ (a repeat of Step 3)

The rate equation for this reaction is:

rate = $k[HBr][O_2]$

(a) Explain which of the above four steps is the rate-determing step for this reaction.

(b) Write the overall equation for the reaction.

11A 5 ACTIVATION ENERGY AND CATALYSIS

SPECIFICATION REFERENCE
11.1(viii) 11.1(ix)
11.11 CP10

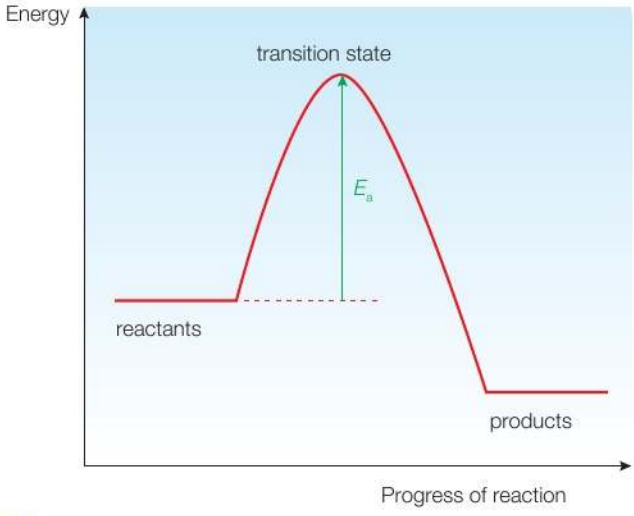
LEARNING OBJECTIVES

- Understand the terms:
  (i) activation energy
  (ii) heterogeneous and homogeneous catalyst
  (iii) autocatalysis.
- Understand the use of a solid (heterogeneous) catalyst for industrial reactions, in the gas phase, in terms of providing a surface for the reaction.

ACTIVATION ENERGY, E_a

In **Topic 9** (**Book 1: IAS**), we defined activation energy, E_a, as the minimum energy that colliding particles must possess for a reaction to occur.

The activation energy represents the energy that the colliding particles must obtain in order to reach the energy level of the *transition state* (see **Section 11A.6** for more details). Once the energy level of the transition state has been reached, the particles can react to form the products and release energy as they do so. The energy profile diagram for an exothermic reaction is shown in **fig A**.



▲ **fig A** A reaction profile for an elementary exothermic reaction.

CATALYSTS

In **Topic 9** (**Book 1: IAS**), we defined a catalyst as a substance that increases the rate of a chemical reaction but is chemically unchanged at the end of the reaction.

We then went on to explain that a catalyst works by providing an alternative route for the reaction, and that this alternative route has a lower activation energy than the original route. Although the original route is still available for the reactants, most collisions resulting in reaction will occur by the alternative route, since the fraction of particles possessing this lower activation energy will be greater.

Catalysts can be divided into two groups:

- homogeneous catalysts
- heterogeneous catalysts.

HOMOGENEOUS CATALYSTS

A *homogeneous catalyst* is in the same phase (solid, liquid, solution or gas) as the reactants.

Many reactions in aqueous solution are catalysed by the hydrogen ion, $H^+(aq)$. An example is the iodination of propanone, a reaction we discussed in detail in **Section 11A.4**:

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

The production of chlorine radicals from chlorofluorocarbons (CFCs) is responsible for the destruction of ozone in the upper atmosphere. Ultraviolet radiation from the Sun produces chlorine radicals, Cl•, from CFCs such as dichlorodifluoromethane, CCl_2F_2:

$$CCl_2F_2(g) \xrightarrow{\text{ultraviolet radiation}} \cdot CClF_2(g) + Cl\cdot(g)$$

The chlorine radicals then take part in a chain reaction with ozone:

$$Cl\cdot(g) + O_3(g) \rightarrow ClO\cdot(g) + O_2(g)$$
$$ClO\cdot(g) + O_3(g) \rightarrow 2O_2(g) + Cl\cdot(g)$$

The chlorine radical is regenerated and so is acting as a catalyst. Since the catalyst is in the same phase, i.e. the gas phase, as the reacting species, it is classified as a homogeneous catalyst.

The overall reaction is:

$$2O_3(g) \rightarrow 3O_2(g)$$

Another reaction involving a homogeneous catalyst is the one between peroxydisulfate ions and iodide ions in aqueous solution:

$$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

This reaction is catalysed by either $Fe^{2+}(aq)$ or $Fe^{3+}(aq)$.

With $Fe^{2+}(aq)$, the reaction mechanism is:

Step 1: $S_2O_8^{2-}(aq) + 2Fe^{2+}(aq) \rightarrow 2SO_4^{2-}(aq) + 2Fe^{3+}(aq)$

Step 2: $2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

With $Fe^{3+}(aq)$ as the catalyst, Steps 1 and 2 occur in the reverse order

LEARNING TIP

The action of $Fe^{2+}(aq)$ or $Fe^{3+}(aq)$ as a catalyst in the reaction between $S_2O_8^{2-}(aq)$ and $I^-(aq)$ is sometimes explained in terms of standard electrode potentials ($E^\ominus$ values).

However, it is important to note that standard electrode potentials can predict only the thermodynamic feasibility of a reaction, and not the kinetics. The reason why the reactions are faster in the presence of either Fe^{2+} or Fe^{3+} is that the activation energies for both Steps 1 and 2 are lower than the activation energy for the overall reaction.

HETEROGENEOUS CATALYSTS

A *heterogeneous* catalyst is in a different phase to that of the reactants.

Two important uses of heterogeneous catalysts in industry are in the Haber process and the Contact process. The use of solid vanadium(V) oxide (V_2O_5), in which the vanadium changes its oxidation number, in the Contact process is described in the section on transition elements (**Topic 17**).

We will now describe the action of solid iron as a catalyst in the reaction between nitrogen gas and hydrogen gas to form ammonia gas in the Haber process.

The equation for the formation of ammonia in the Haber process is:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Iron is able to act as a catalyst because it can form an *interstitial hydride* with hydrogen molecules. In this hydride, hydrogen atoms are held in spaces between the metal ions in the lattice. The atoms are then able to react with nitrogen molecules that are adsorbed onto the metal surface nearby. There are three stages in catalysis involving surface **adsorption**. These are:

1. *Adsorption* – the reactants are first adsorbed onto the surface of the catalyst.
2. *Reaction* – the reactant molecules are held in positions that enable them to react together.
3. *Desorption* – the product molecules leave the surface.

The rate of reaction is controlled by how fast the reactants are adsorbed and how fast the products are desorbed. As mentioned in **Topic 9** (**Book 1: IAS**), once the surface of the iron is covered with molecules, there is no further increase in reaction rate even if the pressure of the reactants is increased.

DID YOU KNOW?

The efficiency of a heterogeneous catalyst depends on the surface of the catalyst. In particular, the efficiency of the catalyst can be affected significantly by poisoning and by the use of promoters.

POISONING

Many catalysts are made ineffective by trace impurities. For example, catalysts used in hydrogenation reactions are poisoned by sulfur impurities. This is one reason why nickel is preferred to platinum as a catalyst in the hydrogenation of alkenes. Nickel is relatively inexpensive, so a large quantity can be used. If some of it is deactivated by poisoning, enough will remain for it still to be effective. On the other hand, platinum is expensive, and so small quantities would have to be used. In this case, it is possible for all of the catalyst to become poisoned.

PROMOTERS

The spacing on the surface of the catalyst is important. For example, only some surfaces of the iron crystals act as effective catalysts in the Haber process. The addition of traces of potassium oxide and aluminium oxide act as promoters by producing *active sites* where the reaction takes place most readily.

DID YOU KNOW?

An interstitial hydride, sometimes called a metallic hydride, is not strictly a compound. It is more like an alloy than a compound. The hydrogen absorbs into the metal and can exist in the form of either atoms or diatomic molecules. For example, palladium absorbs up to 900 times its own volume of hydrogen at room temperature, and forms palladium hydride. This material has been considered as a method to carry hydrogen in fuel cells for use in vehicles (see Section 14.2.1).

A heterogeneous catalyst is used in a three-way catalytic converter, which is employed in the exhaust of cars. This converts unburned hydrocarbons into water and carbon dioxide, and carbon and carbon monoxide into carbon dioxide. It also converts

oxides of nitrogen into oxygen and nitrogen. The reactions involved are discussed in more detail in **Topic 17**.

DID YOU KNOW?

The electrophilic addition reaction between ethene and bromine vapour (**Topic 5 (Book 1: IAS)**) was once thought to be entirely a homogeneous process in which the molecules collided in the gas phase and reacted.

It is now known that the reaction takes place almost entirely on the surface of the walls of the reaction vessel. The reaction takes place very quickly on a glass surface and even more quickly when the surface of the vessel is covered with a polar substance such as cetyl alcohol (hexadecan-1-ol, $CH_3(CH_2)_{15}OH$). It takes place even more quickly still on a surface of stearic acid (octadecanoic acid, $CH_3(CH_2)_{16}COOH$), presumably because stearic acid molecules are more polar than molecules of cetyl alcohol.

However, if the walls of the reaction vessel are coated with a non-polar substance such as paraffin wax, very little reaction, if any, occurs. So, a process once thought to be a homogeneous gas phase reaction is now known to be heterogeneously catalysed.

It is sometimes stated that a catalyst does not change the products of a reaction, but only increases the rate at which the reaction takes place. This is not entirely true, as demonstrated by the different products obtained when hot ethanol vapour is passed over different solid catalysts.

With copper or nickel as the catalyst, at 400 °C, a dehydrogenation reaction takes place in which the products are ethanal and hydrogen:

$$CH_3CH_2OH(g) \longrightarrow CH_3CHO(g) + H_2(g)$$

If, at the same temperature, aluminium oxide is used as the catalyst, the more familiar dehydration reaction into ethene and water takes place:

$$CH_3CH_2OH(g) \longrightarrow CH_2{=}CH_2(g) + H_2O(g)$$

It is interesting to note that the dehydrogenation catalysts, copper and nickel, adsorb hydrogen very strongly, whereas the dehydration catalyst, aluminium oxide, adsorbs water in preference.

These two examples illustrate that there is a lot more to heterogeneous catalysis than is mentioned at International A Level. This might encourage you to do your own research into the topic and learn even more about it.

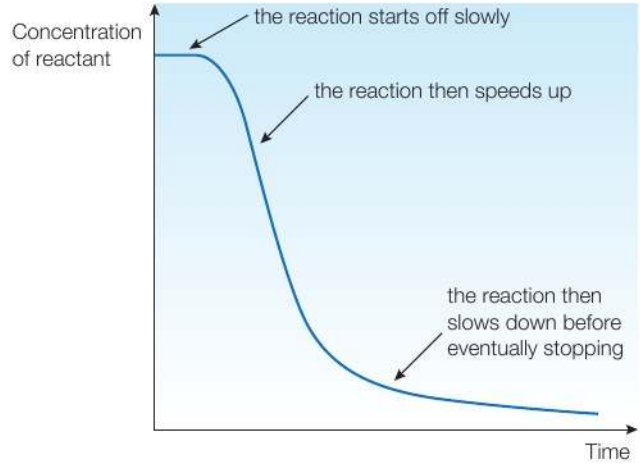
OXIDATION OF ETHANEDIOIC ACID BY MANGANATE(VII) IONS

In **autocatalysis**, the reaction is catalysed by one of its products. One of the simplest examples of this is the oxidation of ethanedioic acid by acidified potassium manganate(VII). The equation for the reaction is:

$$5(COOH)_2(aq) + 2MnO_4^-(aq) + 6H^+(aq) \longrightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

The reaction is very slow at room temperature, but is catalysed by manganese(II) ions, Mn^{2+}. The Mn^{2+} ions are not present initially, so the reaction starts off extremely slowly at room temperature. However, Mn^{2+} is a product of the reaction. As soon as it is produced in a catalytic amount, the reaction rate increases.

You can show this effect by plotting the concentration of one of the reactants against time. The graph obtained is unlike the normal rate curve for a reaction.



▲ **fig B** A graph of concentration against time for an autocatalysed reaction.

In **fig B** you can see the slow (uncatalysed) reaction at the beginning. As catalyst begins to be formed in the mixture, the reaction speeds up – it gets faster and faster as more and more catalyst is formed. Eventually, the rate decreases in the normal way as the concentrations of the reactants decrease.

LEARNING TIP

Do not assume that a rate curve similar to the curve in **fig B** always shows an example of autocatalysis. There are other effects that might produce a similar graph.

For example, if the reaction involves a solid reacting with a liquid, liquid might have to penetrate a substance on the surface of the solid before the expected reaction can happen.

Another possibility is that the reaction is strongly exothermic and the temperature is not being controlled. The heat evolved during the reaction speeds up the reaction.

CHECKPOINT

1. Explain why only a small quantity of a homogeneous catalyst is required in order for it to be effective.
2. Cars in some parts of the world still run on leaded petrol. Leaded petrol contains a compound called tetraethyl lead, $(CH_3CH_2)_4Pb$. Tetraethyl lead reacts in the engine with oxygen and forms lead and lead(II) oxide. These substances remove radical intermediates in the combustion reactions, thus increasing the octane rating of the petrol. Excess lead and lead(II) oxide is removed by adding 1,2-dibromoethane. This forms volatile lead(II) bromide that escapes through the exhaust.

   Suggest why a catalytic converter could not be used in a car running on leaded petrol.

SUBJECT VOCABULARY

adsorption the adhesion of atoms, molecules or ions to the surface of a solid

autocatalysis when a reaction product acts as a catalyst for the reaction

11A 6 EFFECT OF TEMPERATURE ON THE RATE CONSTANT

SPECIFICATION REFERENCE 11.10

LEARNING OBJECTIVES

- Use the Arrhenius equation to explain the effect of temperature on the rate constant of a reaction.
- Use calculations and graphical methods to find the activation energy for a reaction from experimental data.

RELATIONSHIP BETWEEN TEMPERATURE AND RATE OF REACTION

In **Book 1** we discussed qualitatively why an increase in temperature increased the rate of a reaction. There are two reasons for this:

- an increase in the fraction of molecules with energy equal to or greater than the activation energy for the reaction
- an overall increase in the frequency of collisions between the reacting molecules.

The second effect is considerably less significant than the first and means that we can effectively ignore the overall increase in the frequency of collisions.

THE ARRHENIUS EQUATION

EXAM HINT

You will always be given the Arrhenius equation if it is needed in an exam question.

In 1889, Svante Arrhenius, a Swedish chemist, proposed a quantitative relationship between temperature and the rate constant, k, for a reaction. This is described by the *Arrhenius equation* and is usually expressed in the form:

$$k = Ae^{\left(-\frac{E_a}{RT}\right)}$$

where:

- A is a constant known as the pre-exponential factor, which is a measure of the rate at which collisions occur irrespective of their energy. It also includes other factors, the most important of which is that reactions can only occur when the molecules are correctly orientated at the time of collision.
- E_a is the activation energy of the reaction.
- R is the gas constant.
- T is the absolute temperature (i.e. the temperature in kelvin).

If we take natural logarithms (i.e. logarithms to the base 'e') of the Arrhenius equation. we obtain:

$$\ln k = -\frac{E_a}{R}\frac{1}{T} + \ln A$$

If a graph of $\ln k$ is plotted against $\frac{1}{T}$, a straight line is obtained with a gradient of $-\frac{E_a}{R}$

This provides an experimental method for determining the activation energy of a reaction.

The intercept with the vertical axis gives $\ln A$.

WORKED EXAMPLE

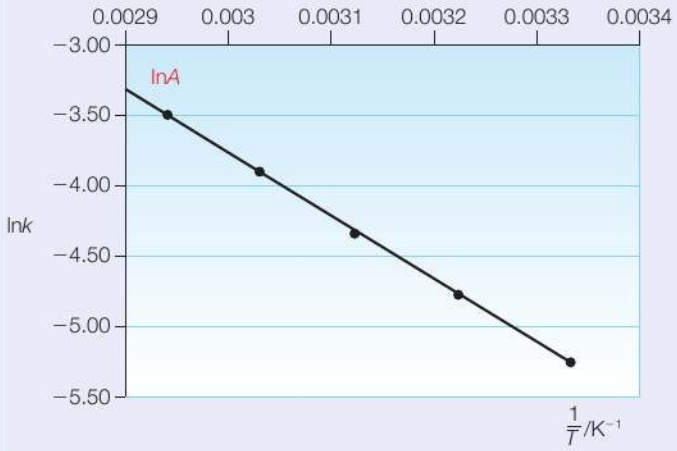
Table A shows some data related to the reaction between peroxydisulfate and iodide ions in aqueous solution:

$$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

TEMPERATURE/K	MAGNITUDE OF RATE CONSTANT, k	$\ln k$	$1/T/K^{-1}$
300	0.00513	−5.27	0.00333
310	0.00833	−4.79	0.00323
320	0.0128	−4.36	0.00313
330	0.0201	−3.91	0.00303
340	0.0301	−3.50	0.00294

table A

Fig A shows ln*k* plotted against $\frac{1}{T}$.



▲ **fig A** A graph of ln*k* against 1/*T*.

The gradient of the line, $-\frac{E_a}{R}$

$= -\frac{(-3.50 - -5.27)}{(0.00333 - 0.00294)}$

$= -4538$

[$R = 8.31\ J\ mol^{-1}\ K^{-1}$]

$E_a = -(-4538 \times 8.31) = 37\,700\ J\ mol^{-1} = +37.7\ kJ\ mol^{-1}$ (to 3 s.f.)

DID YOU KNOW?

When we state that the relationship between ln*k* and $-\frac{1}{T}$ is linear, we have assumed that both E_a and *A* remain constant over a range of temperatures. This is not strictly true. However, the change in the values of E_a and *A* with temperature are insignificant compared with the effect of temperature on the rate constant, and so can be ignored.

DID YOU KNOW?

The significance of the factor $e^{-\frac{E_a}{RT}}$

The factor $e^{-\frac{E_a}{RT}}$ represents the fraction of collisions that have energy equal to or greater than the activation energy, E_a. For example, a reaction with an E_a of 60 kJ mol^{-1} gives a fraction, at 298 K, of:

$$e^{-(60\,000/8.31 \times 298)} = 3 \times 10^{-11}$$

So, only one collision in 3×10^{11} has sufficient energy to react.

It is often assumed that the fraction of collisions with energy equal to or greater than E_a is the same as the fraction of molecules with this energy. This is not strictly true, but the difference is insignificant at high energies, where the fraction is very small.

It is, therefore, reasonable to draw the molecular energy distribution curve, instead of the collision frequency curve, when demonstrating the effect of temperature on collision frequency. (See **Topic 9** (**Book 1: IAS**).

LEARNING TIP

The effect of E_a on reaction rate:

- reactions with a large E_a are slow, but the rate increases rapidly with an increase in temperature
- reactions with a small E_a are fast, but the rate does not increase as rapidly with an increase in temperature
- catalysed reactions have small values of E_a.

DID YOU KNOW?

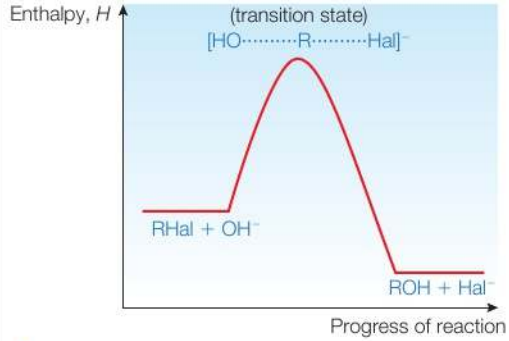
Reaction profiles

When drawing reaction profile diagrams for multi-step reactions, it is important to distinguish between an *intermediate* and a *transition state*:

- an intermediate has an energy minimum
- a transition state occurs at the top of the energy curve and therefore has an energy maximum.

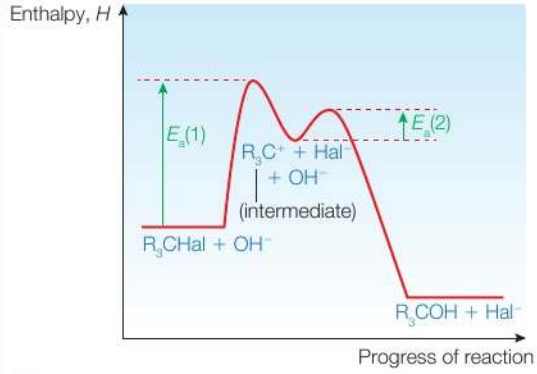
An intermediate is a definite chemical species that exists for a finite length of time. By comparison, a transition state has no significant permanent lifetime of its own: it exists for a period of the order of 10^{-15} seconds, when the molecules are in contact with one another. Even a very reactive intermediate, with a lifetime of only 10^{-6} seconds, has a long lifetime in comparison with the period that colliding molecules are in contact with one another.

A simple one-step reaction, such as the S_N2 hydrolysis of a primary halogenoalkane, has a single maximum energy. The reaction profile for the hydrolysis of a primary halogenoalkane, represented as RHal, is therefore:



▲ **fig B** The reaction profile diagram for the hydrolysis of a primary halogenoalkane.

The reaction profile for the S_N1 hydrolysis of a tertiary halogenoalkane, R_3CHal, which is a two-step reaction involving the intermediate R_3C^+, is:



▲ **fig C** The reaction profile diagram for the hydrolysis of a tertiary halogenoalkane.

LEARNING TIP

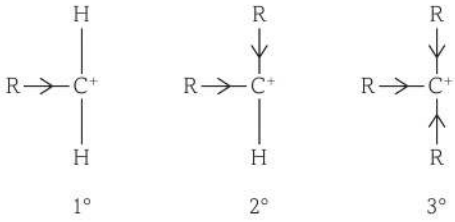
In the S_N1 hydrolysis of a tertiary halogenoalkane, the first step of the mechanism is the rate-determining step of the reaction, and it therefore has the higher activation energy. That is, $E_a(1) > E_a(2)$.

If the second step in a reaction mechanism is rate-determining, $E_a(1) < E_a(2)$.

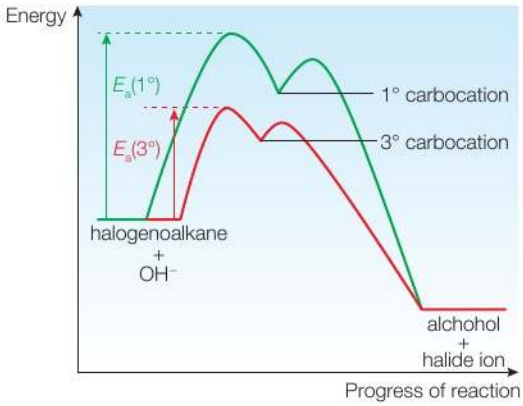
DID YOU KNOW?

The relative rates of S_N1 and S_N2 reactions depend on structure

You learned in **Book 1** that alkyl groups donate electrons by the inductive effect. This means that the stability of carbocations increases in the order 1° < 2° < 3°, as the number of alkyl groups donating electrons towards the positive carbon atom increases.



As the stability of the carbocation increases, the activation energy for the reaction leading to its formation also decreases.



▲ **fig D** Reaction profiles for the S_N1 hydrolysis of primary and tertiary halogenoalkanes.

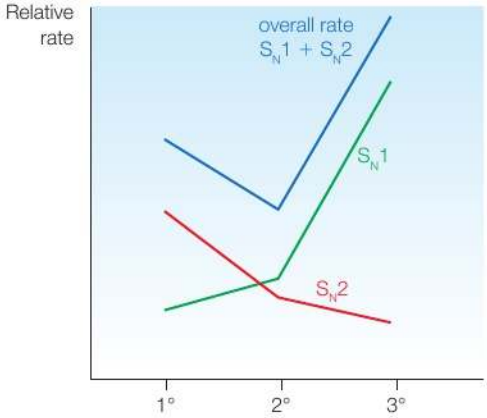
We therefore expect the rate of the S_N1 reaction to increase in the order 1° < 2° < 3°.

No carbocations are formed during the S_N2 reaction. In an S_N2 reaction, the transition state has five groups arranged around the central carbon atom. It is therefore more crowded than either the starting halogenoalkane or the alcohol product, each of which have only four groups around the central carbon atom.

Alkyl groups are much larger than hydrogen atoms. Therefore, the more alkyl groups around the central carbon atom, the more crowded will be the transition state, and the higher the activation energy for its formation.

We therefore expect the rate of the S_N2 reaction to increase in the order 3° < 2° < 1°.

The two effects reinforce one another. The S_N1 reaction is fastest with tertiary halides and slowest with primary halides, while the S_N2 reaction is fastest with primary halides and slowest with tertiary halides. Overall, primary halides react predominantly via the S_N2 mechanism and tertiary halides react predominantly via the S_N1 mechanism. Secondary halides tend to react via a mixture of the two mechanisms.



▲ **fig E** S_N1 and S_N2 hydrolysis reactions for primary, secondary and tertiary halogenoalkanes.

CHECKPOINT

SKILLS CREATIVITY

1. Use the Arrhenius equation to explain why an increase in temperature results in an increase in the rate of reaction.
2. The rate constant, k, for a reaction increases from 10.0 s^{-1} to 100.0 s^{-1} when the temperature is increased from 300 K to 400 K.

   Calculate the activation energy, E_a, for this reaction. [R = 8.31 J mol^{-1} K^{-1}].